



Integration Formulas for JEE, Class 12th, and Class 11th

Integration is one of the most fundamental and challenging topics in calculus, forming a crucial component of JEE Main and Advanced mathematics syllabus. This comprehensive guide presents all essential integration formulas, methods, and techniques required for success in competitive examinations and board exams.

Integration Formulas Reference

Category	Formula	Result
Basic Formulas	$\int 1 \, dx$	$x + C$
Basic Formulas	$\int k \, dx$	$kx + C$
Basic Formulas	$\int x^n \, dx \, (n \neq -1)$	$x^{n+1}/(n+1) + C$
Basic Formulas	$\int (1/x) \, dx$	$\ln x + C$
Exponential/Logarithmic	$\int e^x \, dx$	$e^x + C$
Exponential/Logarithmic	$\int a^x \, dx$	$a^x/\ln(a) + C$
Exponential/Logarithmic	$\int \ln(x) \, dx$	$x \ln(x) - x + C$
Trigonometric	$\int \sin x \, dx$	$-\cos x + C$
Trigonometric	$\int \cos x \, dx$	$\sin x + C$
Trigonometric	$\int \tan x \, dx$	$\ln \sec x + C$
Trigonometric	$\int \sec x \, dx$	$\ln \sec x + \tan x + C$
Trigonometric	$\int \sec^2 x \, dx$	$\tan x + C$
Trigonometric	$\int \csc^2 x \, dx$	$-\cot x + C$

Essential Integration Formulas for JEE, Class 11th & 12th

Basic Integration Formulas

Elementary Integration Results

The foundation of integration lies in understanding these fundamental formulas derived directly from differentiation:[\[1\]](#) [\[2\]](#) [\[3\]](#)

Power Rule and Basic Functions:

- $\int 1, dx = x + C$
- $\int k, dx = kx + C$ (where k is constant)
- $\int x^n, dx = \frac{x^{n+1}}{n+1} + C$ ($n \neq -1$)
- $\int \frac{1}{x}, dx = \ln|x| + C$

Exponential and Logarithmic Functions:

- $\int e^x, dx = e^x + C$
- $\int a^x, dx = \frac{a^x}{\ln a} + C$ ($a > 0, a \neq 1$)
- $\int \ln x, dx = x \ln x - x + C$

Fundamental Theorems of Integration

These basic theorems form the framework for all integration techniques:^{[1] [4]}

1. **Linearity Property:** $\int [f(x) \pm g(x)], dx = \int f(x), dx \pm \int g(x), dx$
2. **Constant Multiple Rule:** $\int k \cdot f(x), dx = k \int f(x), dx$
3. **Inverse Relationship:** $\frac{d}{dx} \left[\int f(x), dx \right] = f(x)$

FORMULAE FOR INTEGRAL CALCULUS

Compiled By: Pawan Sir [Maths Expert]

1. Standard Formulae Of Integrals:

$$A. \int x^n dx = \frac{x^{n+1}}{n+1} + k, n \neq -1$$

$$B. \int \frac{1}{x} dx = \log|x| + k$$

$$C. \int a^x dx = a^x \frac{1}{\log a} + k$$

$$D. \int e^x dx = e^x + k \text{ or } \int e^{ax} dx = \frac{1}{a} e^{ax} + k$$

$$E. \int \sin x dx = -\cos x + k \text{ \& } \int \sin(ax) dx = -\frac{1}{a} \cos(ax) + k \quad O. \int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1}x + k = -\operatorname{cosec}^{-1}x + k$$

$$F. \int \cos x dx = \sin x + k \text{ \& } \int \cos(ax) dx = \frac{1}{a} \sin(ax) + k$$

$$G. \int \tan x dx = \log|\sec x| + k$$

$$\text{OR} = -\log|\cos x| + k$$

$$H. \int \cot x dx = \log|\sin x| + k$$

$$\text{OR} = -\log|\operatorname{cosec} x| + k$$

$$I. \int \sec x dx = \log|\sec x + \tan x| + k$$

$$\text{OR} = \log\left|\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right| + k$$

$$J. \int \operatorname{cosec} x dx = \log|\operatorname{cosec} x - \cot x| + k$$

$$\text{OR} = \log\left|\tan\frac{x}{2}\right| + k$$

$$V. \int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log|x + \sqrt{x^2 - a^2}| + k$$

$$W. \int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log|x + \sqrt{x^2 + a^2}| + k$$

$$X. \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + k$$

$$Y. \int \frac{1}{ax+b} dx = \frac{1}{a} \log|ax+b| + k, \text{ where 'a' is any constant (and obviously, k is the integral constant).}$$

$$Z. \int \lambda dx = \lambda x + k, \text{ where } \lambda \text{ is a constant (and, k is the integral constant).}$$

$$K. \int \sec^2 x dx = \tan x + k$$

$$L. \int \operatorname{cosec}^2 x dx = -\cot x + k$$

$$M. \int \sec x \cdot \tan x dx = \sec x + k$$

$$N. \int \operatorname{cosec} x \cdot \cot x dx = -\operatorname{cosec} x + k$$

$$P. \int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log\left|\frac{x-a}{x+a}\right| + k.$$

$$Q. \int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \log\left|\frac{a+x}{a-x}\right| + k$$

$$R. \int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + k$$

$$\text{OR } \int \frac{1}{1+x^2} dx = \tan^{-1}x + k = -\cot^{-1}x + k$$

$$S. \int \frac{1}{\sqrt{x^2 - a^2}} dx = \log|x + \sqrt{x^2 - a^2}| + k$$

$$T. \int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + k$$

$$\text{OR } \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}x + k = -\cos^{-1}x + k$$

$$U. \int \frac{1}{\sqrt{x^2 + a^2}} dx = \log|x + \sqrt{x^2 + a^2}| + k$$

2. Methods of integration:

Though there is no general method for finding the integral of a function, yet here we have considered the following Methods based on observations for evaluating the integral of a function:

a) Integration by Substitution Method –

In this method we change the integral $\int f(x) dx$, into another form by changing the independent variable x to t by substituting $x = g(t)$.

Consider

$$I = \int f(x) dx$$

$$\text{Put } x = g(t) \text{ so that } \frac{dx}{dt} = g'(t).$$

Then $dx = g'(t) dt$

$$\text{i.e., } I = \int f(x) dx = \int f[g(t)] g'(t) dt.$$

Trigonometric Integration Formulas

Standard Trigonometric Integrals

These formulas are frequently tested in JEE and form the backbone of trigonometric integration:
[\[5\]](#) [\[6\]](#) [\[7\]](#)

Basic Trigonometric Functions:

- $\int \sin x, dx = -\cos x + C$
- $\int \cos x, dx = \sin x + C$
- $\int \tan x, dx = \ln|\sec x| + C = -\ln|\cos x| + C$
- $\int \cot x, dx = \ln|\sin x| + C$
- $\int \sec x, dx = \ln|\sec x + \tan x| + C$
- $\int \csc x, dx = \ln|\csc x - \cot x| + C$

Squared Trigonometric Functions:

- $\int \sec^2 x, dx = \tan x + C$
- $\int \csc^2 x, dx = -\cot x + C$
- $\int \sec x \tan x, dx = \sec x + C$
- $\int \csc x \cot x, dx = -\csc x + C$

Advanced Trigonometric Integration

For integrals of the form $\int \sin^m x \cos^n x, dx$: [\[4\]](#)

Strategy Selection:

- If m is odd: substitute $\cos x = t$
- If n is odd: substitute $\sin x = t$
- If both are odd: choose substitution based on higher power
- If both are even: use power reduction formulas

Power Reduction Formulas:

- $\sin^2 x = \frac{1 - \cos 2x}{2}$
- $\cos^2 x = \frac{1 + \cos 2x}{2}$

I	Inverse Trigonometric Functions	$\sin^{-1} x ; \cos^{-1} x ; \tan^{-1} x \dots$
L	Logarithmic Functions	$\log x$
A	Algebraic Functions	$1 ; x ; x^2 + 1 ; \dots$
T	Trigonometric Functions	$\sin x ; \cos x ; \tan x \dots$
E	Exponential Functions	e^x

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The ILATE rule chart for selecting functions in integration by parts, categorizing functions from Inverse Trigonometric to Exponential.

Integration by Parts

The ILATE/LIATE Rule

Integration by parts is governed by the formula: $\int u \, dv = uv - \int v \, du$ ^{[8] [9] [10] [11]}

Function Priority (ILATE/LIATE):

- Inverse trigonometric functions
- Logarithmic functions

- Algebraic functions
- Trigonometric functions
- Exponential functions

The function appearing first in this hierarchy should be chosen as 'u'.

Special Integration by Parts Formulas

Advanced Results:

- $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$
- $\int [xf'(x) + f(x)] dx = xf(x) + C$

Standard Results from Integration by Parts:

- $\int e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \sin(bx) - b \cos(bx)) + C$
- $\int e^{ax} \cos(bx) dx = \frac{e^{ax}}{a^2 + b^2} (b \sin(bx) + a \cos(bx)) + C$

$$\int_I x \sin 3x dx$$

Integrating by parts,

$$= x \left\{ \int \sin 3x dx \right\} - \int \left\{ \frac{d}{dx} (x) \int \sin 3x dx \right\} dx$$

$$= -x \frac{\cos 3x}{3} - \int 1 \left\{ -\frac{\cos 3x}{3} \right\} dx$$

$$= -\frac{1}{3} x \cos 3x + \frac{1}{3} \int \cos 3x dx$$

$$= -\frac{1}{3} x \cos 3x + \frac{1}{9} \sin 3x + C$$

ILATE rule diagram illustrating the priority order for choosing functions in integration by parts.

Integration by Substitution

Integration Methods for JEE Prep

Method	Technique	When to Use	Example
Direct Integration	Standard Formulas	When integrand matches standard forms	$\int x^n dx$, $\int \sin x dx$, $\int e^x dx$
Substitution	$u = g(x)$, $du = g'(x)dx$	When derivative of inner function is present	$\int f(g(x))g'(x)dx$
Integration by Parts	$\int u dv = uv - \int v du$	Product of two different types (ILATE)	$\int x e^x dx$, $\int x \sin x dx$
Trigonometric Substitution	$x = a \sin \theta$, $a \tan \theta$, $a \sec \theta$	$\sqrt{a^2-x^2}$, $\sqrt{x^2+a^2}$, $\sqrt{x^2-a^2}$	$\int \frac{1}{\sqrt{a^2-x^2}} dx$
Partial Fractions	Decompose rational functions	Rational functions $P(x)/Q(x)$	$\int \frac{x+1}{(x-1)(x+2)} dx$
Trigonometric Integration	Power reduction, identities	$\int \sin^n x \cos^n x dx$	$\int \sin^2 x \cos^3 x dx$
Rational Function Method	Complete the square	Quadratic denominators	$\int \frac{1}{ax^2+bx+c} dx$
Special Exponential	Multiply by e^{-x}/e^x	$\int \frac{1}{1+e^x} dx$ type	$\int \frac{e^x}{1+e^{2x}} dx$
Reduction Formula	Express I_n in terms of I_{n-1}	High powers of functions	$\int \sin^n x dx$, $\int \tan^n x dx$

Integration Methods and Techniques for JEE Preparation

Standard Substitution Types

Function of Function Form:

If the integral is $\int f[g(x)] \cdot g'(x) dx$, substitute $g(x) = t$ ^[12] ^[13] ^[14]

Product of Function and Derivative:

For $\int f^n(x) \cdot f'(x) dx$, substitute $f(x) = t$

Standard Results:

- $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$
- $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)} + C$
- $\int f^n(x) \cdot f'(x) dx = \frac{f^{n+1}(x)}{n+1} + C$ ($n \neq -1$)

Trigonometric Substitutions

For expressions involving square roots: ^[15] ^[14]

- $\sqrt{a^2 - x^2}$: substitute $x = a \sin \theta$
- $\sqrt{x^2 + a^2}$: substitute $x = a \tan \theta$
- $\sqrt{x^2 - a^2}$: substitute $x = a \sec \theta$

Special Substitutions:

- For $\sqrt{(x-a)(b-x)}$: substitute $x = a \cos^2 \theta + b \sin^2 \theta$
- For $\frac{1}{ax + b}$: substitute $ax + b = t$

Standard Integral Formulas

Inverse Trigonometric Forms

These formulas are essential for JEE Advanced problems:^[16]

- $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$

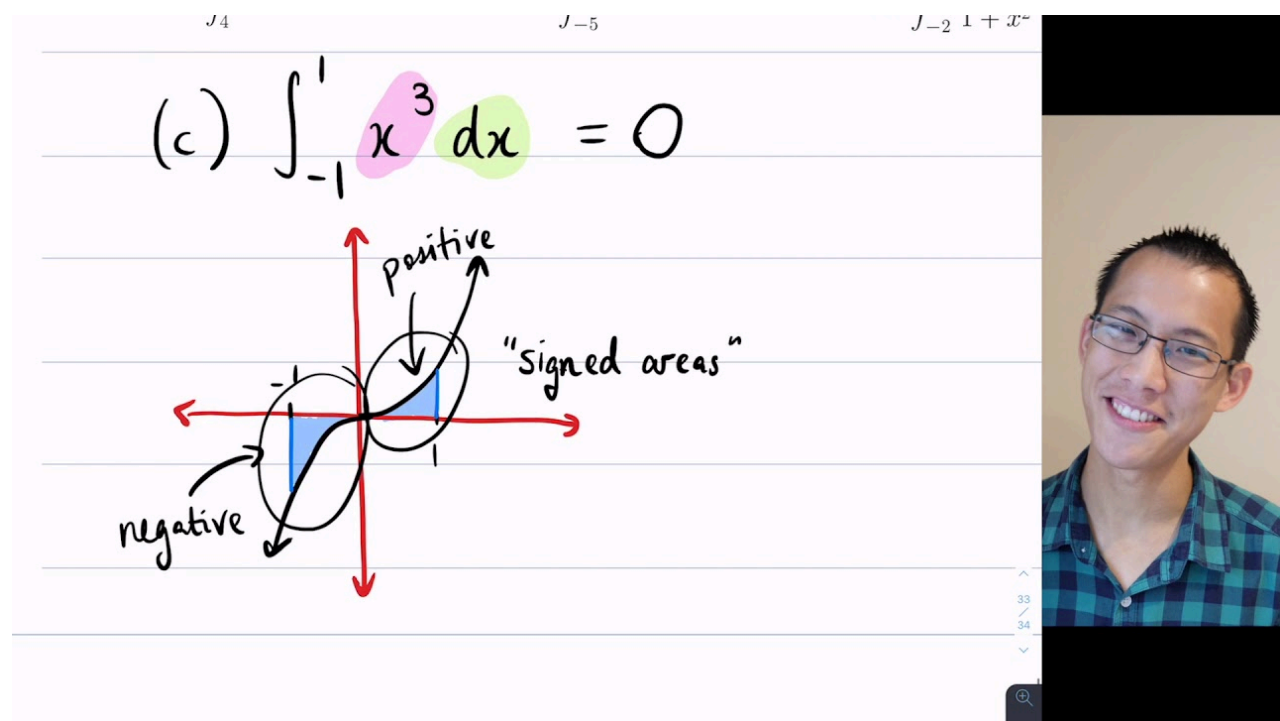
Advanced Standard Forms

Involving Square Roots:

- $\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln|x + \sqrt{x^2 + a^2}| + C$
- $\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln|x + \sqrt{x^2 - a^2}| + C$
- $\int \sqrt{x^2 + a^2} dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\ln|x + \sqrt{x^2 + a^2}| + C$

Rational Functions:

- $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln\left|\frac{x-a}{x+a}\right| + C$
- $\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln\left|\frac{a+x}{a-x}\right| + C$



Graphical representation and evaluation of the integral of x cubed from -1 to 1 showing signed areas and symmetry leading to zero.

Definite Integration

Properties of Definite Integrals

Fundamental Properties: [\[17\]](#) [\[18\]](#) [\[19\]](#) [\[20\]](#)

- $\int_a^b f(x) \, dx = -\int_b^a f(x) \, dx$
- $\int_a^a f(x) \, dx = 0$
- $\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$
- $\int_a^b f(x) \, dx = \int_a^b f(t) \, dt$

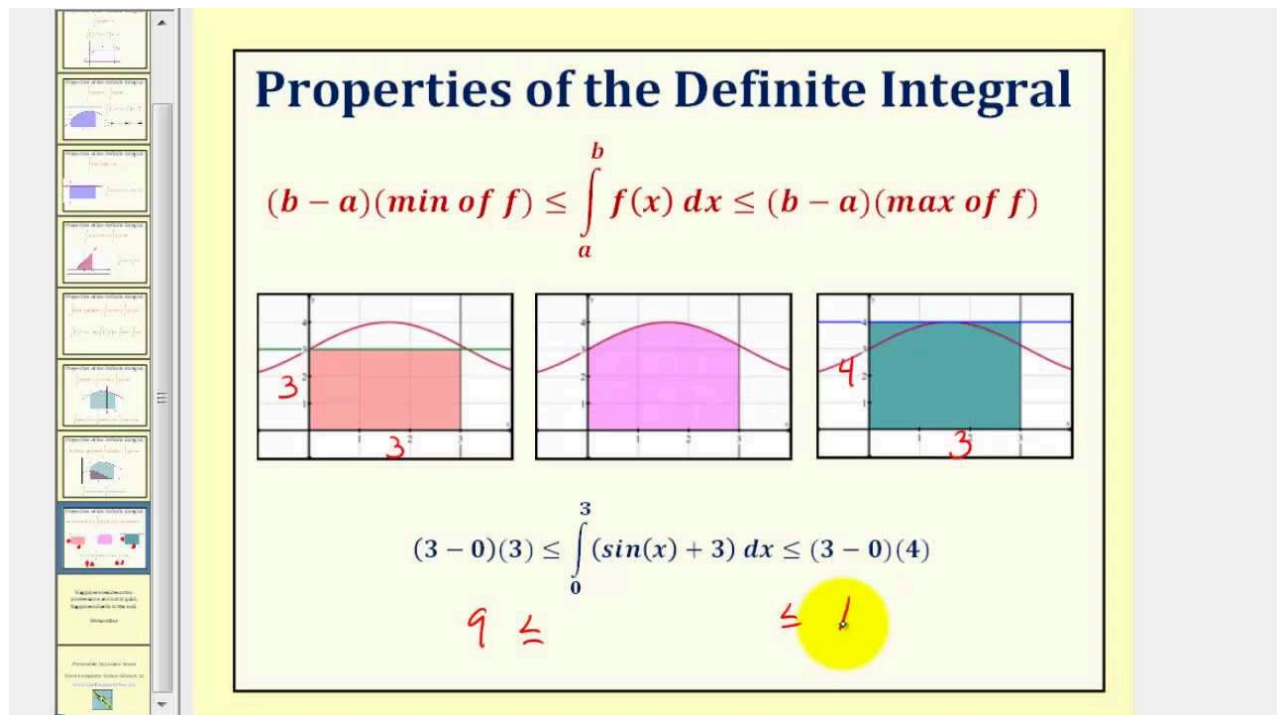
Special Properties:

- $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$
- $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$ if $f(x)$ is even
- $\int_{-a}^a f(x) \, dx = 0$ if $f(x)$ is odd

Definite Integration by Parts

Formula: $\int_a^b u \, dv = [uv]_a^b - \int_a^b v \, du$

Where $[uv]_a^b = u(b)v(b) - u(a)v(a)$



Properties of definite integrals with illustrative graphs and an example using sine function showing bounds on the integral.

Reduction Formulas

Trigonometric Reduction Formulas

For Powers of Sine and Cosine: ^[21] ^[22] ^[23] ^[24]

- $\int \sin^n x, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x, dx$
- $\int \cos^n x, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x, dx$

For Tangent and Secant:

- $\int \tan^n x, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x, dx$
- $\int \sec^n x, dx = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{n-2}{n-1} \int \sec^{n-2} x, dx$

Exponential and Logarithmic Reduction

For Exponential Functions:

- $\int x^n e^{mx}, dx = \frac{1}{m} x^n e^{mx} - \frac{n}{m} \int x^{n-1} e^{mx}, dx$

For Logarithmic Functions:

- $\int (\ln x)^n, dx = x(\ln x)^n - n \int (\ln x)^{n-1}, dx$

Special Integration Techniques

Rational Function Integration

Using Partial Fractions: ^[4]

For proper rational functions $\frac{P(x)}{Q(x)}$:

1. **Non-repeated linear factors:** $\frac{1}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$
2. **Repeated linear factors:** $\frac{1}{(x-a)^n} = \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \dots + \frac{A_n}{(x-a)^n}$
3. **Quadratic factors:** Include terms like $\frac{Ax+B}{ax^2+bx+c}$

Integration of Irrational Functions

Complete the Square Method:

For integrals involving $\sqrt{ax^2 + bx + c}$, express as:

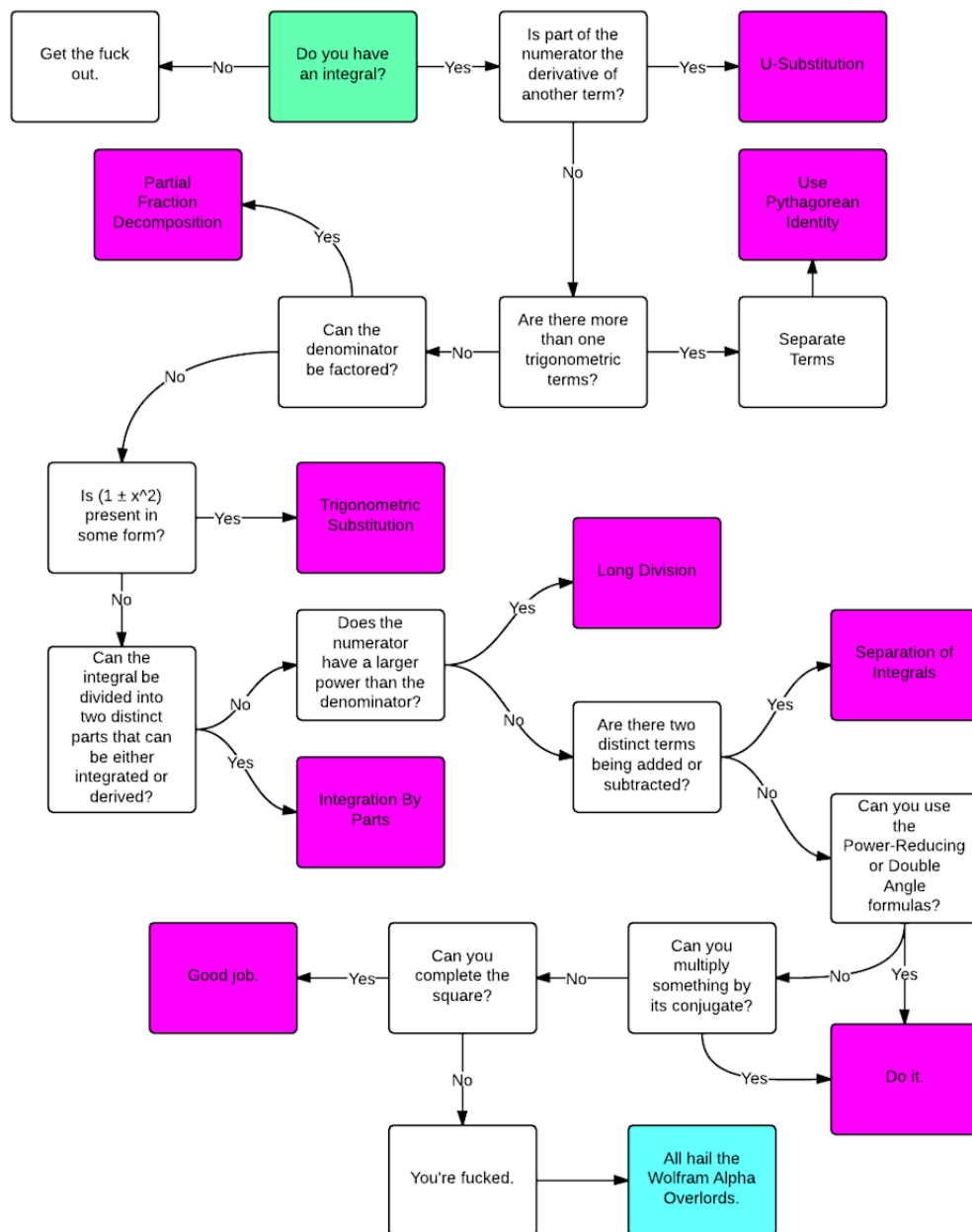
$$ax^2 + bx + c = a \left[\left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a^2} \right]$$

Special Exponential Integration

Common Types: ^[4]

- $\int \frac{e^x}{1+e^x}, dx$: substitute $e^x = t$
- $\int \frac{1}{1+e^x}, dx$: multiply by $\frac{e^{-x}}{e^{-x}}$

- $\int \frac{e^x}{e^x - 1} dx$, dx : substitute $e^x - 1 = t$



Flowchart for choosing integration techniques including substitution, partial fractions, trigonometric substitution, and more.

Problem-Solving Strategy Guide

Choosing the Right Method

Decision Framework:

1. **Direct Integration:** Check if integrand matches standard forms
2. **Substitution:** Look for function-derivative pairs or composite functions
3. **Parts:** Use ILATE rule for products of different function types
4. **Partial Fractions:** For rational functions where degree of numerator < denominator
5. **Trigonometric Substitution:** For expressions involving $\sqrt{a^2 \pm x^2}$

Common JEE Problem Types

High-Frequency Topics:

- Integration by substitution with trigonometric functions
- Integration by parts with logarithmic and inverse trigonometric functions
- Definite integration using properties and substitution
- Integration of rational functions using partial fractions
- Application of reduction formulas for higher powers

Tips for JEE Success

1. **Master the ILATE rule** - it's crucial for integration by parts problems
2. **Practice standard substitutions** - they appear frequently in competitive exams
3. **Memorize reduction formulas** - essential for complex trigonometric integrals
4. **Understand definite integration properties** - they provide shortcuts in evaluation
5. **Work with partial fractions systematically** - break down complex rational functions step by step

Conclusion

Integration is a vast topic requiring systematic practice and thorough understanding of various techniques. The formulas and methods presented in this guide form the complete foundation needed for JEE Main and Advanced success. Regular practice with different problem types, combined with a solid understanding of when to apply each technique, will lead to mastery of this crucial mathematical skill.

Remember that integration often requires combining multiple techniques, so developing the ability to recognize patterns and choose appropriate methods is as important as memorizing the formulas themselves. Focus on understanding the underlying principles rather than rote memorization, and always verify your answers through differentiation when possible.

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